Physics - Graduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. When an increase in speed doubles the momentum of a moving body its kinetic energy
   - (a) decreases slightly
   - (b) increases but does not reach double
   - (c) remains the same
   - (d) halves
   - (e) quadruples
2. The flight of a blimp best illustrates
   - (a) Archimedes' principle
   - (b) Pascal's principle
   - (c) Newton's first law
   - (d) The law of conservation of energy
   - (e) Hooke's law
3. How do astronomers think Jupiter generates its internal heat?
   - (a) the gravitational pull from its moons
   - (b) heat produced by its magnetic field
   - (c) by contracting changing gravitational potential energy into thermal energy
   - (d) heat absorption from the Sun
   - (e) nuclear fusion in the core
4. An ultrasonic transducer used for medical diagnosis oscillates at 6.7 Mhz.How long does each oscillation take, and what is the angular frequency? (Unit: 10^7 rad/s)
   - (a) 5.3
   - (b) 1.5
   - (c) 4.2
   - (d) 8.4
   - (e) 3.8
5. Why are there no impact craters on the surface of Io?
   - (a) Any craters that existed have been eroded through the strong winds on Io's surface.
   - (b) Io is protected by a layer of gas that prevents any objects from reaching its surface.

(c) The gravitational pull of nearby moons prevent objects from hitting Io.

(d) Io's magnetic field deflects incoming asteroids, preventing impact craters.

(e) Io did have impact craters but they have all been buried in lava flows.

True / False (10 marks)

Answer True or False

6. True or False: An aircraft can produce a sonic boom while flying at a constant speed below the sp
7. True or False: The angular acceleration of the turntable is 0.58 sec^-2, indicating a slower increa
8. True or False: The binding energy of the 4He nucleus is approximately 27.5 MeV, suggesting a w
9. True or False: A particle moving at v=0.60c will decay after traveling only 100 m in the lab frame
10. True or False: Mars' axial tilt is less pronounced than Earth's, contributing to its more extreme se

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the principles of thin-film interference and calculate the required thickness of an MgF coating on a glass surface to minimize reflection at a wavelength of 550 nm.
12. Discuss the implications of relativistic length contraction as two spaceships approach Earth at equal speeds from opposite directions, given that a meterstick on one is measured to be 60 cm by an occupant of the other. What is the speed of each spaceship relative to the observer on Earth, and how does this scenario illustrate the principles of special relativity?

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